

“CHIP DESIGN IS ONE OF THE MOST COMPLEX ENGINEERING DOMAINS IMAGINABLE, AND THERE IS HUGE OPPORTUNITY FOR AI—PARTICULARLY AN AGENTIC APPROACH—TO TRANSFORM EVERYTHING FROM LOGIC DESIGN TO BOARD VALIDATION.”

— **GOKUL V. SUBRAMANIAM, INTEL INDIA PRESIDENT (IN A RECENT INTERVIEW WITH THE TIMES OF INDIA)**



providers to notify and support users with special needs when there is an outage. Most companies struggle with that, but a utility provider in the UK recently tackled this issue using AI agents.

- **Telecom.** Telecom infrastructure and service providers across the world have started using agentic AI for scheduling maintenance, monitoring the infrastructure to pre-empt outages, improving efficiency and resilience, reducing costs, and upping customer service in terms of response times, resolution quality, and more. In India, top internet service providers (ISPs) like ACT Fibernet, GTPL, HFCL, Excitel, Asianet, and Netplus have embraced agentic AI-driven network intelligence solutions, such as Aprecomm’s, to streamline operations.

With access to the network, user analytics, and insights needed to monitor each user’s quality of experience, AI agents identify potential performance issues such as interference and adjust the Wi-Fi parameters to reduce downtime through proactive self-healing and self-optimisation. This enhances user experience and reduces operational costs. “AI Agents are integral to our solutions, functioning as autonomous entities that continuously monitor, analyse, and optimise network performance in real time. Embedded within network devices, these agents can detect anomalies, predict potential failures, and implement corrective

actions proactively without human intervention. This autonomy allows networks to adapt dynamically to changing conditions and user demands, leading to enhanced reliability and performance,” says Pramod Gummaraj, founder and CEO of Aprecomm.

- **Engineering.** Agentic AI can play a significant role in many cutting-edge engineering fields, like chip design. Electronic design automation (EDA) vendors like Synopsis, as well as chipmakers like Intel, have started trialling agentic AI. “Chip design is one of the most complex engineering domains imaginable, and there is huge opportunity for AI—particularly an agentic approach—to transform everything from logic design to board validation,” said Gokul Subramaniam, Intel India president and VP of the client computing group at Intel, in a recent interview with the *Times of India*. Engineering teams at Intel India are developing AI systems to automate chip design.

Subramaniam explained that each stage in semiconductor

engineering, such as design, development, testing, validation and deployment, has its own workflow with numerous tools and people involved. “We believe an agentic approach—thinking of them as sophisticated assistants—can help with tasks like generating code, running simulations, or assisting with placement (of transistors), all while engineers retain full oversight,” he said.

Intel is not building LLMs from the ground up, but using tools like LangChain or LangGraph, and layering their domain-specific knowledge on top. They are training the agents using historical data collected at Intel over 50+ years. Subramaniam highlighted that their engineers are encouraged to document their insights, not just for their benefit but also to feed the AI agents.

He suggested that if the fabless companies mushrooming in India can adopt AI-driven workflows, they could get to market faster and with fewer design spins, which could significantly lower barriers to entry.

The path ahead

With its abilities to autonomously plan and execute the steps needed to achieve a goal, work in a chain of command, seamlessly integrate with existing systems and human employees, learn from its experiences, and constantly improve, agentic AI has proven to be a

“AI IS DOING A GREAT JOB ELIMINATING THE ACCIDENTAL COMPLEXITY. HUMANS ARE STILL NEEDED TO DEAL WITH THE ESSENTIAL COMPLEXITY. AI CAN MAKE MINCEMEAT OF PATTERNS ALREADY DISCOVERED (BY HUMANS). CAN IT FIND TOTALLY NEW PATTERNS... I DON’T KNOW IF AI CAN DO THIS. I DON’T KNOW IF THAT CAN BE BRUTE-FORCED.”

— **SRIDHAR VEMBU, FOUNDER & FORMER CEO, ZOHU CORPORATION (IN A POST ON X)**

